

Store Database Schema

Each block below (table name, description, and attributes) is meant to be copied as one chunk.

Table: users

Description: People who can log in: customers, staff, admins

- id (BIGSERIAL): Primary key
- full_name (VARCHAR(150)): User's full name
- email (VARCHAR(150)): Unique login email
- password_hash (VARCHAR(255)): Hashed password, never stored in plain text
- phone (VARCHAR(30)): Contact phone number
- role (VARCHAR(20)): Access level: customer, admin, or staff
- is_active (BOOLEAN): Whether the account is currently enabled
- created_at (TIMESTAMPTZ): Account creation timestamp
- updated_at (TIMESTAMPTZ): Last profile update timestamp

Table: addresses

Description: Shipping/billing addresses linked to a user

- id (BIGSERIAL): Primary key
- user_id (BIGINT (FK -> users.id)): Owner of this address
- label (VARCHAR(50)): Address label, e.g. Home, Work
- street (VARCHAR(255)): Street address
- city (VARCHAR(100)): City name
- region (VARCHAR(100)): State/province/delivery zone
- country (VARCHAR(100)): Country name
- postal_code (VARCHAR(20)): Postal/ZIP code
- is_default (BOOLEAN): Whether this is the user's default address
- created_at (TIMESTAMPTZ): Address creation timestamp

Table: categories

Description: Product categories, supports nested subcategories via parent_id

- id (BIGSERIAL): Primary key
- name (VARCHAR(120)): Category name
- description (TEXT): Category description
- parent_id (BIGINT (FK -> categories.id)): Parent category id, NULL if top-level
- created_at (TIMESTAMPTZ): Category creation timestamp

Table: products

Description: Sellable items in the store catalog

- id (BIGSERIAL): Primary key
- category_id (BIGINT (FK -> categories.id)): Category this product belongs to
- sku (VARCHAR(64)): Stock keeping unit, unique product code
- name (VARCHAR(200)): Product name
- description (TEXT): Product description
- price (NUMERIC(12,2)): Sale price shown to customers
- cost (NUMERIC(12,2)): Cost to acquire/produce the item, used for margin calculations
- stock_quantity (INTEGER): Units currently available in inventory
- is_active (BOOLEAN): Whether the product is currently listed for sale
- created_at (TIMESTAMPTZ): Product creation timestamp
- updated_at (TIMESTAMPTZ): Last product update timestamp

Table: product_images

Description: Images associated with a product

- id (BIGSERIAL): Primary key
- product_id (BIGINT (FK -> products.id)): Product this image belongs to
- url (VARCHAR(500)): Image URL
- is_primary (BOOLEAN): Whether this is the main product image
- sort_order (INTEGER): Display order among a product's images

Table: orders

Description: Customer purchase transactions

- id (BIGSERIAL): Primary key
- user_id (BIGINT (FK -> users.id)): Customer who placed the order
- address_id (BIGINT (FK -> addresses.id)): Delivery address for the order
- status (VARCHAR(30)): Order lifecycle state: pending, paid, shipped, delivered, cancelled
- total_amount (NUMERIC(12,2)): Final amount charged, including shipping/tax
- placed_at (TIMESTAMPTZ): When the order was placed
- updated_at (TIMESTAMPTZ): Last order status update timestamp

Table: order_items

Description: Individual line items within an order

- id (BIGSERIAL): Primary key
- order_id (BIGINT (FK -> orders.id)): Order this line item belongs to
- product_id (BIGINT (FK -> products.id)): Product purchased
- quantity (INTEGER): Units purchased
- unit_price (NUMERIC(12,2)): Price of the product at time of purchase

Table: reviews

Description: Customer product reviews and ratings

- id (BIGSERIAL): Primary key
- product_id (BIGINT (FK -> products.id)): Product being reviewed
- user_id (BIGINT (FK -> users.id)): Customer who left the review
- rating (SMALLINT (1-5)): Star rating given by the customer
- comment (TEXT): Free-text review comment
- created_at (TIMESTAMPTZ): When the review was submitted

Table: payments

Description: Payment transactions linked to orders

- id (BIGSERIAL): Primary key
- order_id (BIGINT (FK -> orders.id)): Order this payment covers
- method (VARCHAR(30)): Payment method used: card, cash_on_delivery, wallet
- amount (NUMERIC(12,2)): Amount paid
- status (VARCHAR(20)): Payment state: pending, completed, failed, refunded
- paid_at (TIMESTAMPTZ): When payment was completed